

Computer Science & IT



Data Structure

Tree

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



Topic

General tree

Topic

forest

Topic

k-ary tree

Topic

$$L = I(k-1) + 1$$

Topic

Minimum No. of Nodes k^{h+1}

Maximum: $\frac{k^{h+1} - 1}{k - 1}$

Topics to be Covered



Topic

Binary tree

Topic

Height and No. of Nodes

Topic

Binary tree Theorem

Topic

counting of binary tree

Topic



Topic : Tree

Binary tree is a tree in which each node is

having 0, 1 or 2 children. (at most 2 children)

$$n_0 = 2$$

$$n_2 = 1$$

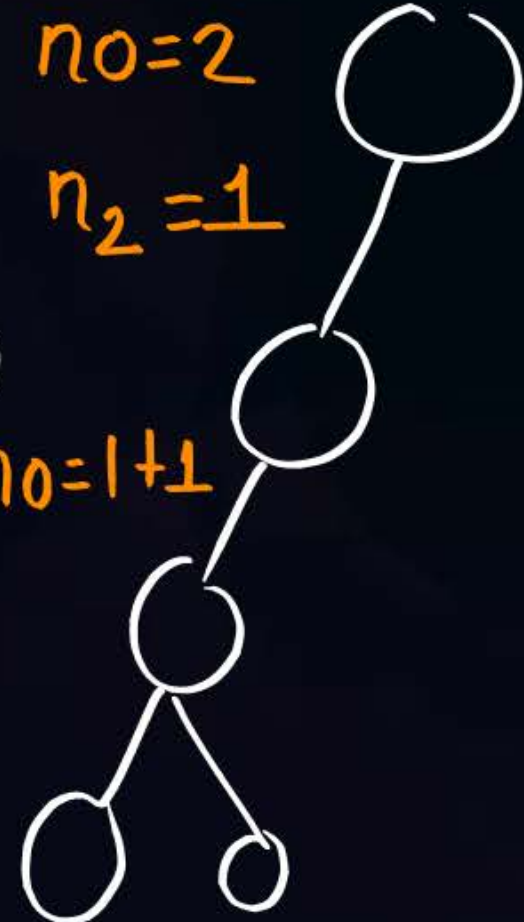
$$n_0 = 1 + 1$$



$$n_0 = 2$$

$$n_2 = 1$$

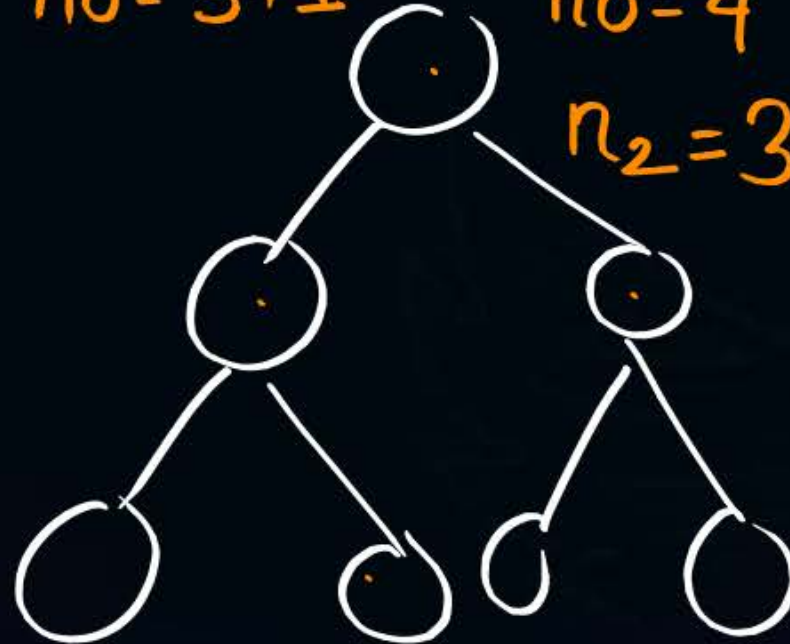
$$n_0 = 1 + 1$$



$$n_0 = 3 + 1$$

$$n_0 = 4$$

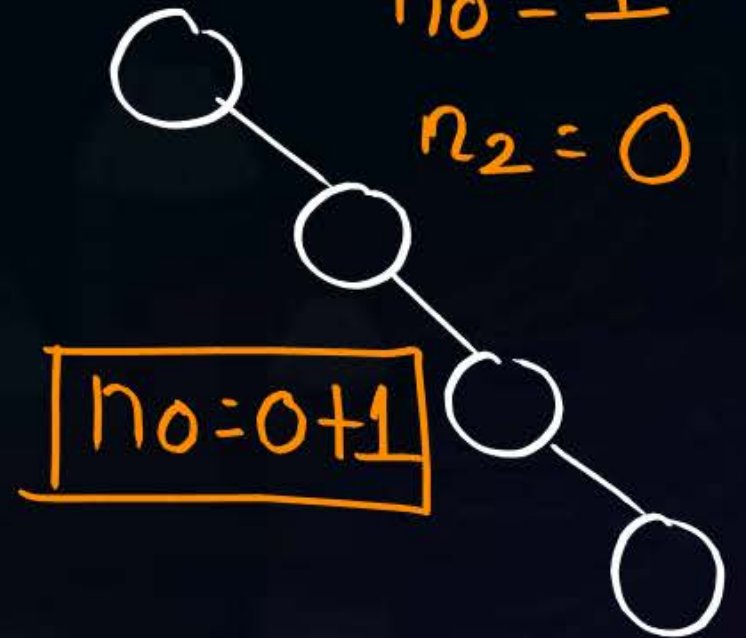
$$n_2 = 3$$



$$n_0 = 1$$

$$n_2 = 0$$

$$n_0 = 0 + 1$$





Topic : Tree



In any binary tree if n_0 is No. of leaves

n_1 is No. of node with 1 children, n_2 is No. of nodes with 2 children.

$$n_0 = n_2 + 1$$

No. of leaves = No. of node with 2 children + 1



Topic : Tree



Binomy tree operation depends upon height of tree

Best case : h - Minimum

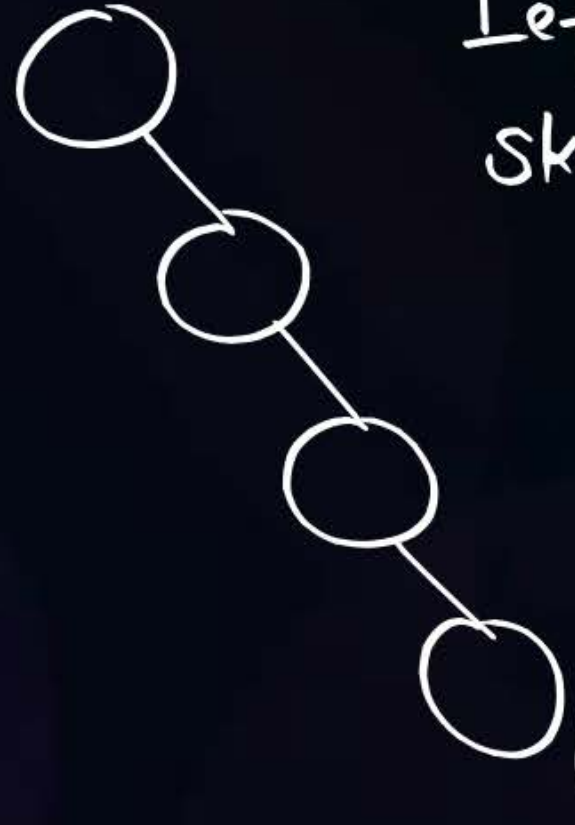
Worst case : h - maximum



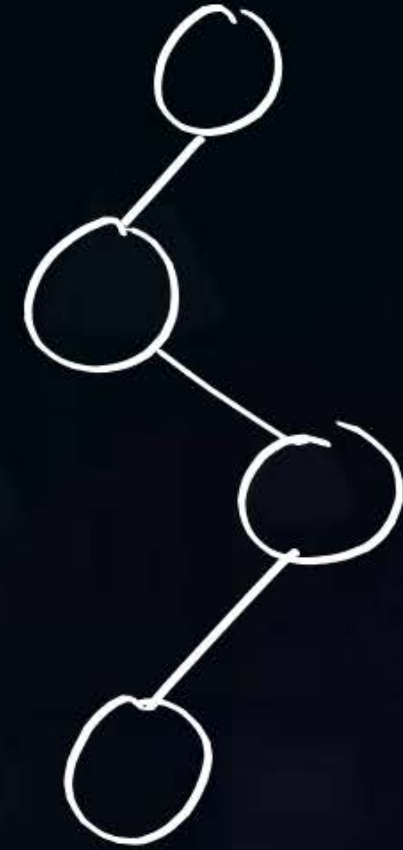
Topic : Skewed Binary Tree

Skewed binary tree: A binary tree in which each node having 1 child except 1 node (Leaf Node)

Right
Skewed

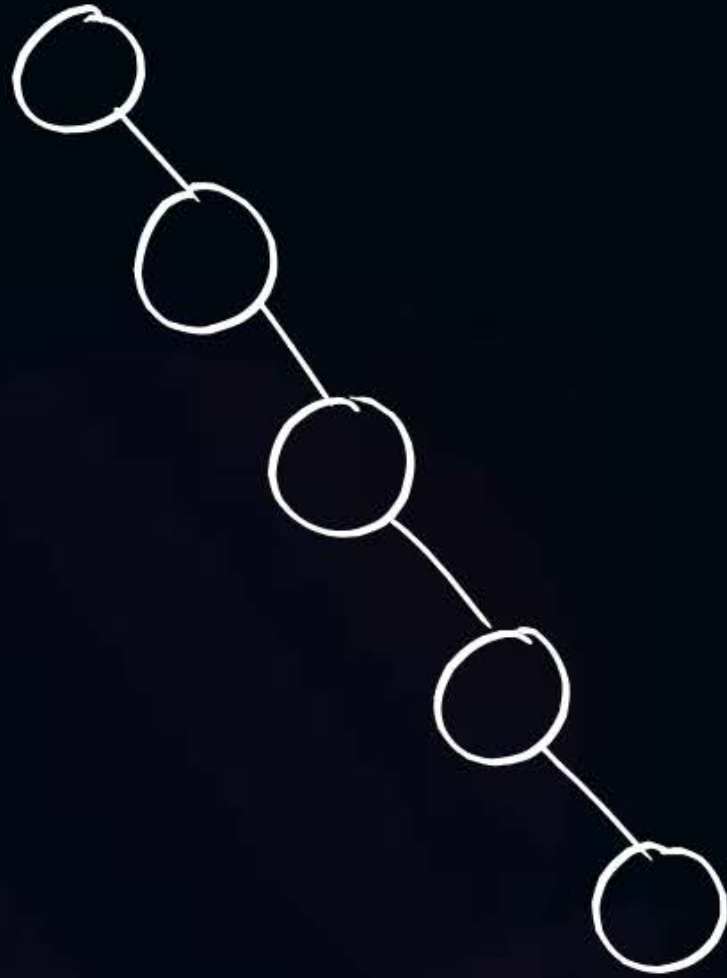


Left
skewed





Topic : Tree



$$n = h + 1$$

$$h = n - 1$$

Any operation on skewed
tree $O(n)$

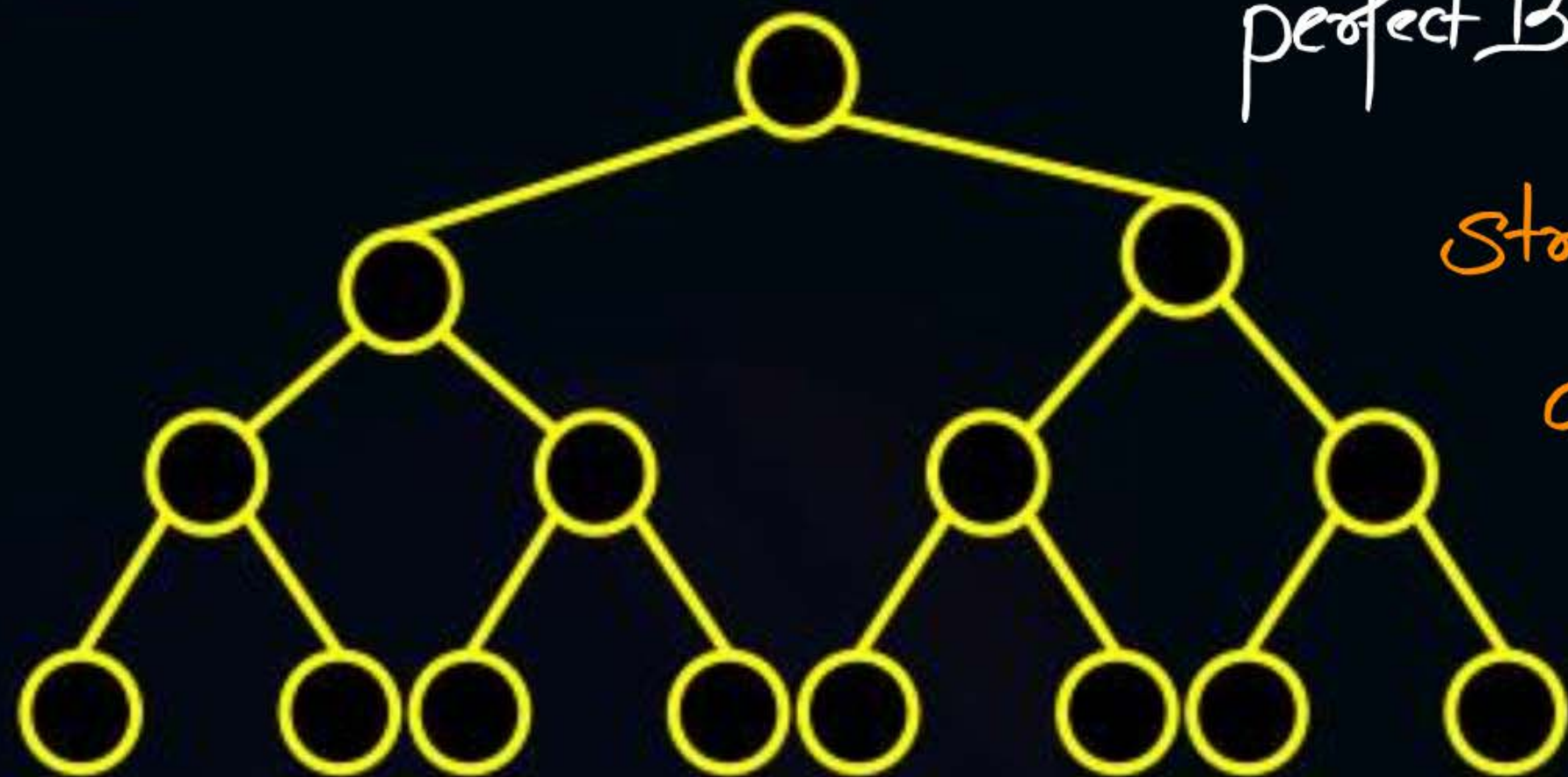


Full Binary Tree



← In some Gate question

perfect Binary tree ← online reference



Structure: Every Level is full
and all the leaves at same height

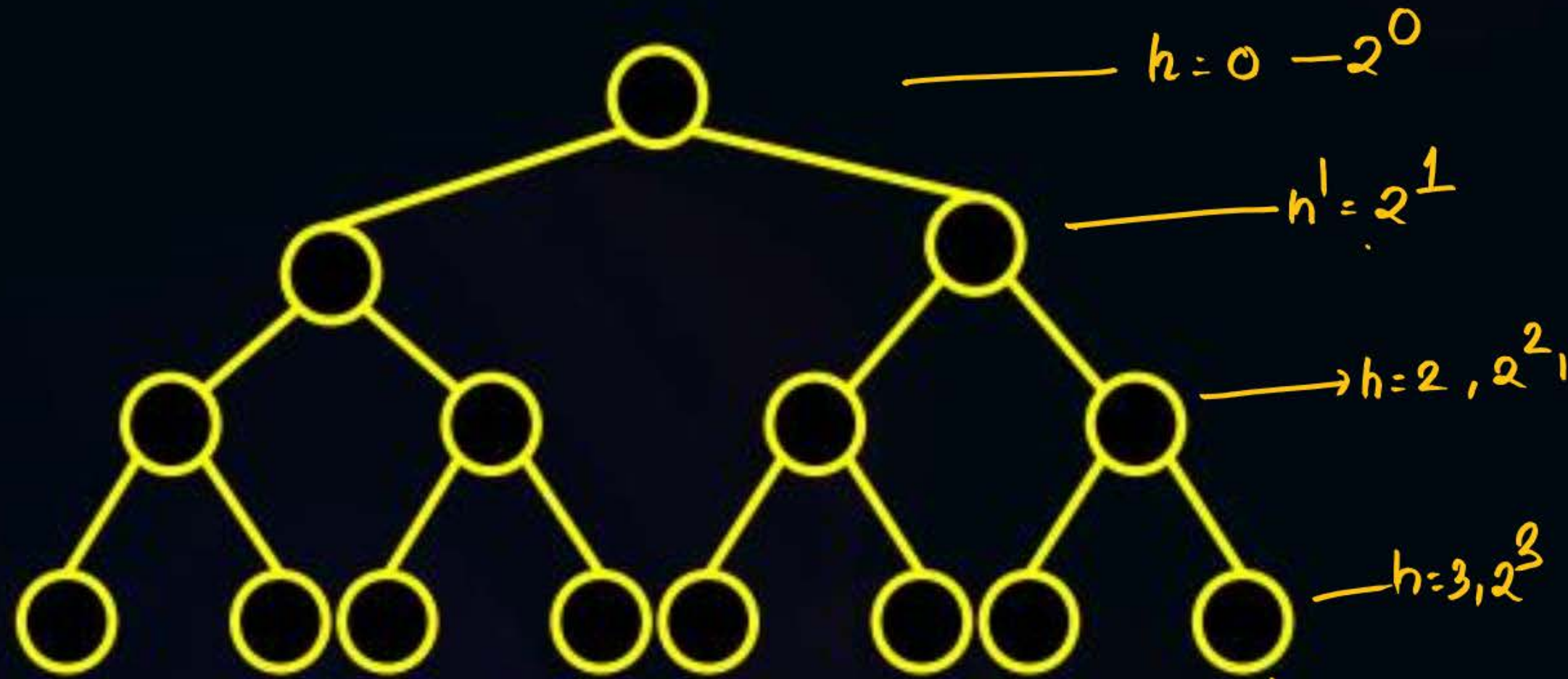
$h=3$, Maximum No. of
Nodes, Minimum height
with n nodes

2-ary ✓

CBT ✓

Minimum Height of binary Tree with N Nodes

FBT-



$$\lfloor \log_2 15 \rfloor = 3$$

Total No. of nodes

$$n = 2^0 + 2^1 + 2^2 + \dots + 2^h$$

$$n = \frac{2^{h+1} - 1}{2 - 1}$$

$$n = 2^{h+1} - 1$$

$$2^{h+1} = n + 1$$

$$2^h = \frac{n+1}{2} \quad (\log_2 \text{ both side})$$

$$h = \log_2 \left(\frac{n+1}{2} \right)$$

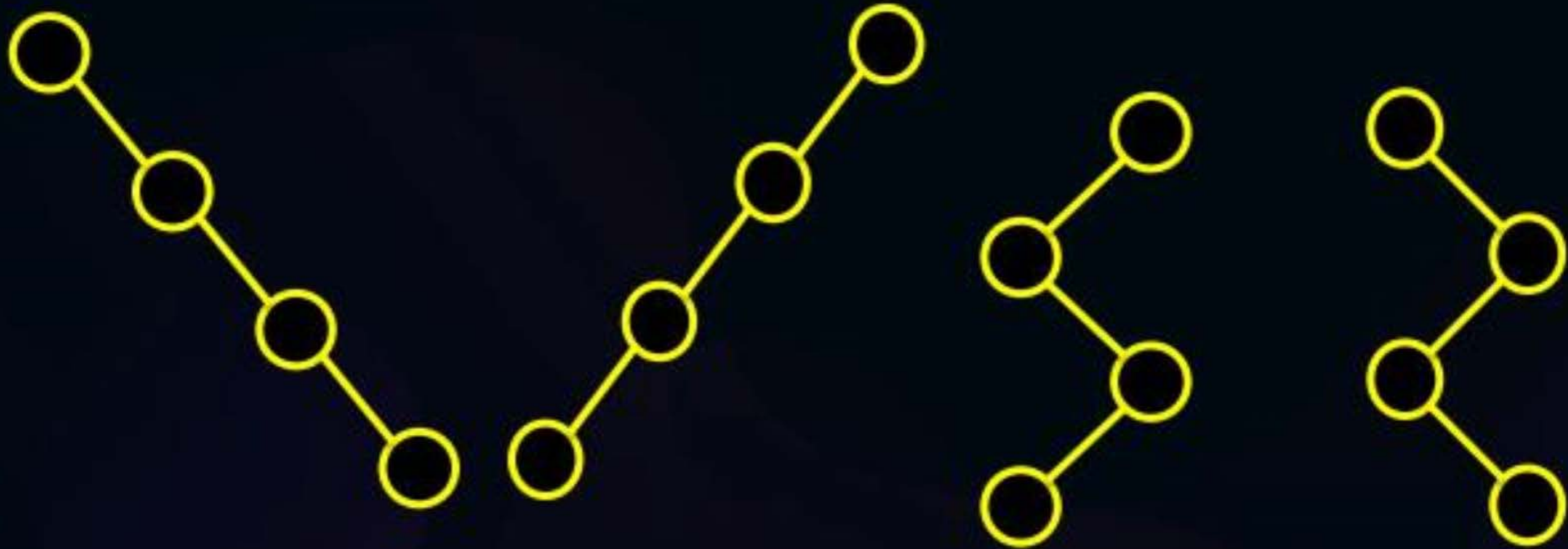
height of binary tree $\left\{ \begin{array}{l} \text{Max: } n-1 \\ \text{Min: } \lfloor \log_2 n \rfloor \end{array} \right.$



Topic : Skewed Binary Tree



A skew tree is a binary tree such that all the nodes except leaf node have *one and only one child*.



Minimum Height of full binary Tree n nodes



Complete Binary tree

in CBT every level is full

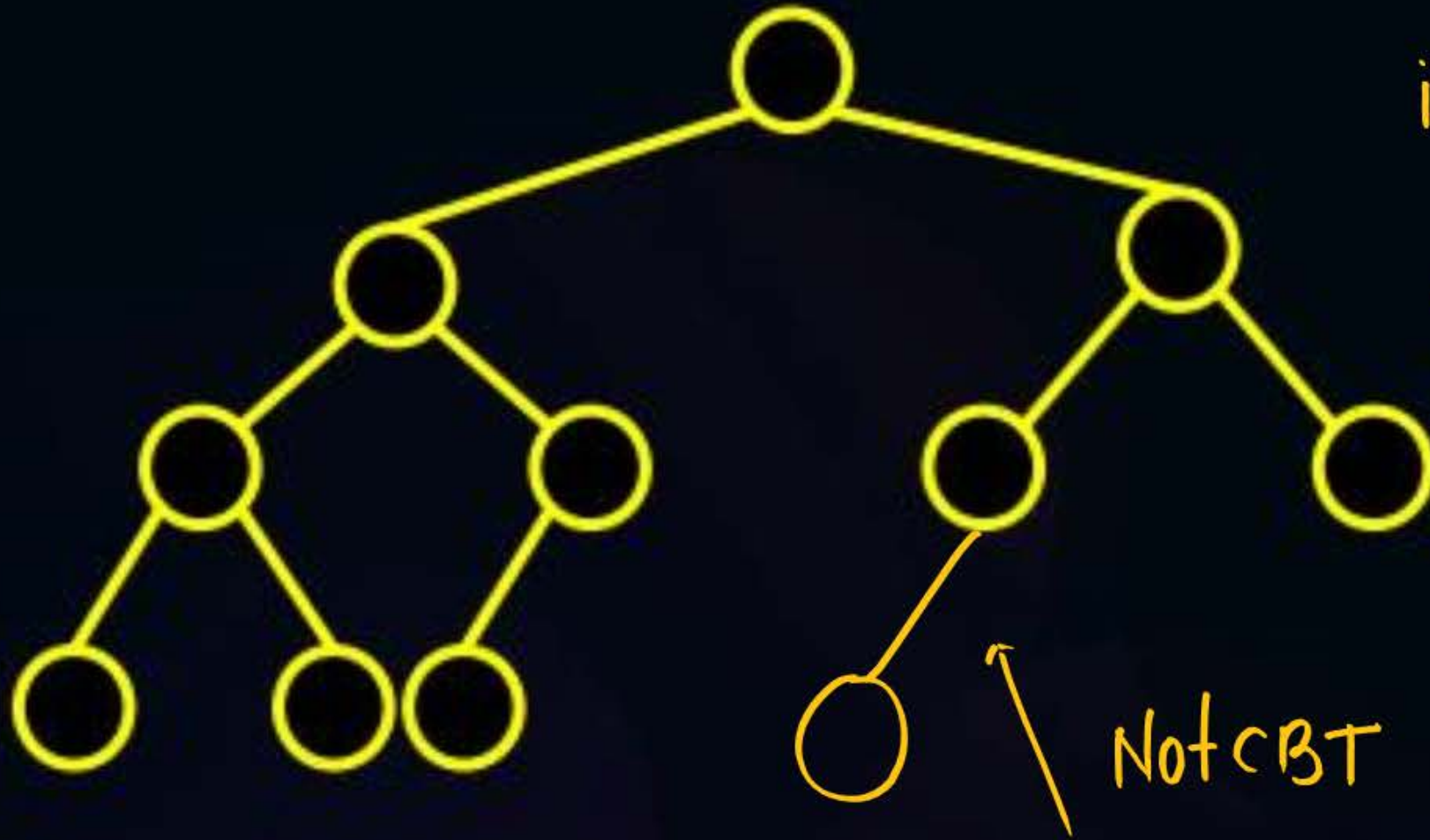
except possibly the last.

In Last Level nodes arranged

strictly left to right

$$\text{height of CBT} = \lfloor \log_2 n \rfloor$$

Minimum Height of full binary Tree n nodes



Complete Binary tree

in CBT every level is full

except possibly the last.

In Last Level nodes arranged

strictly left to right

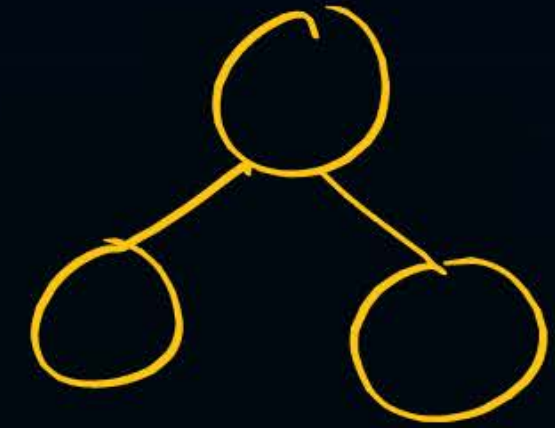
Not CBT



Topic : Counting of unlabeled Tree

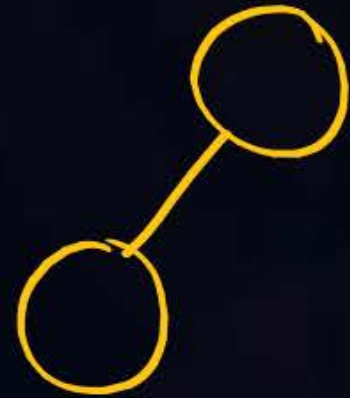
$T(n)$ No. of unlabeled trees with n -nodes

(I) $n=1$



↑ unlabeled tree

(II) $n=2$



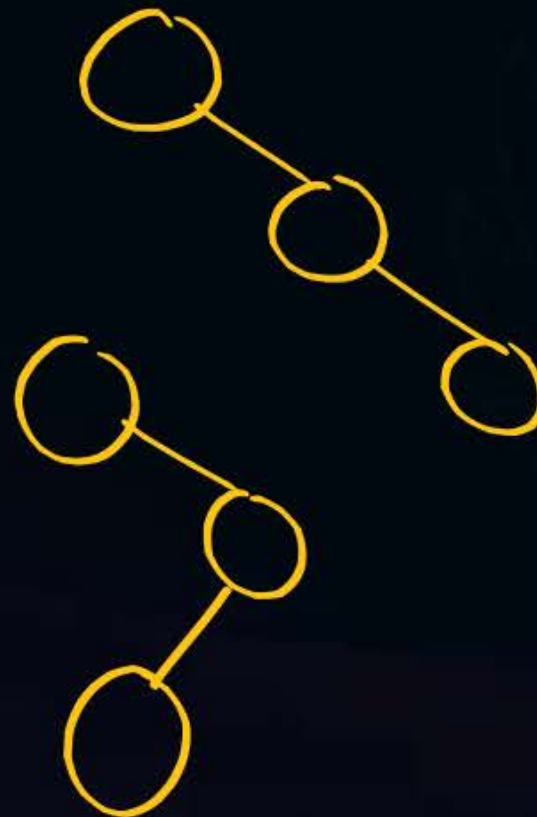
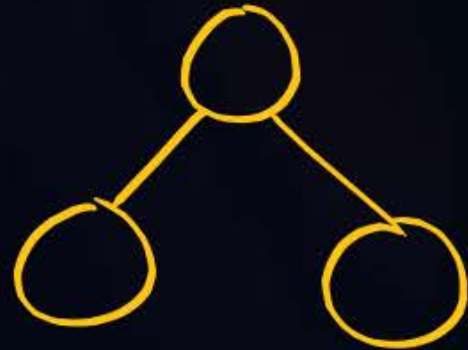
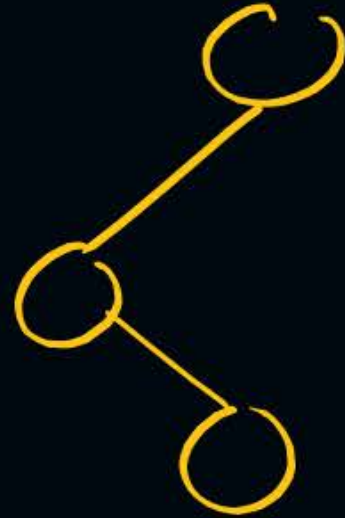
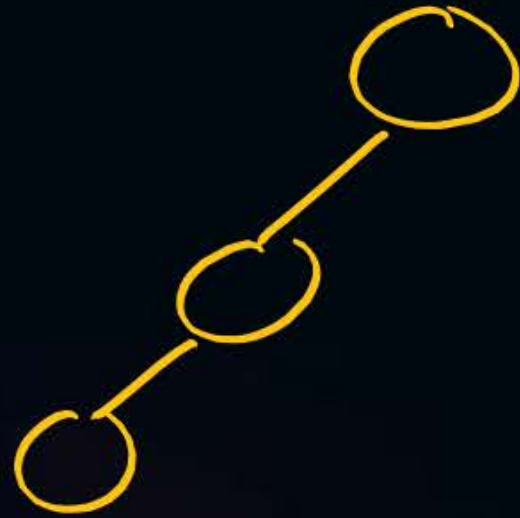
or





Topic : Counting of unlabeled Tree

(III) $n=3$



2

L

R

2

0

1

1

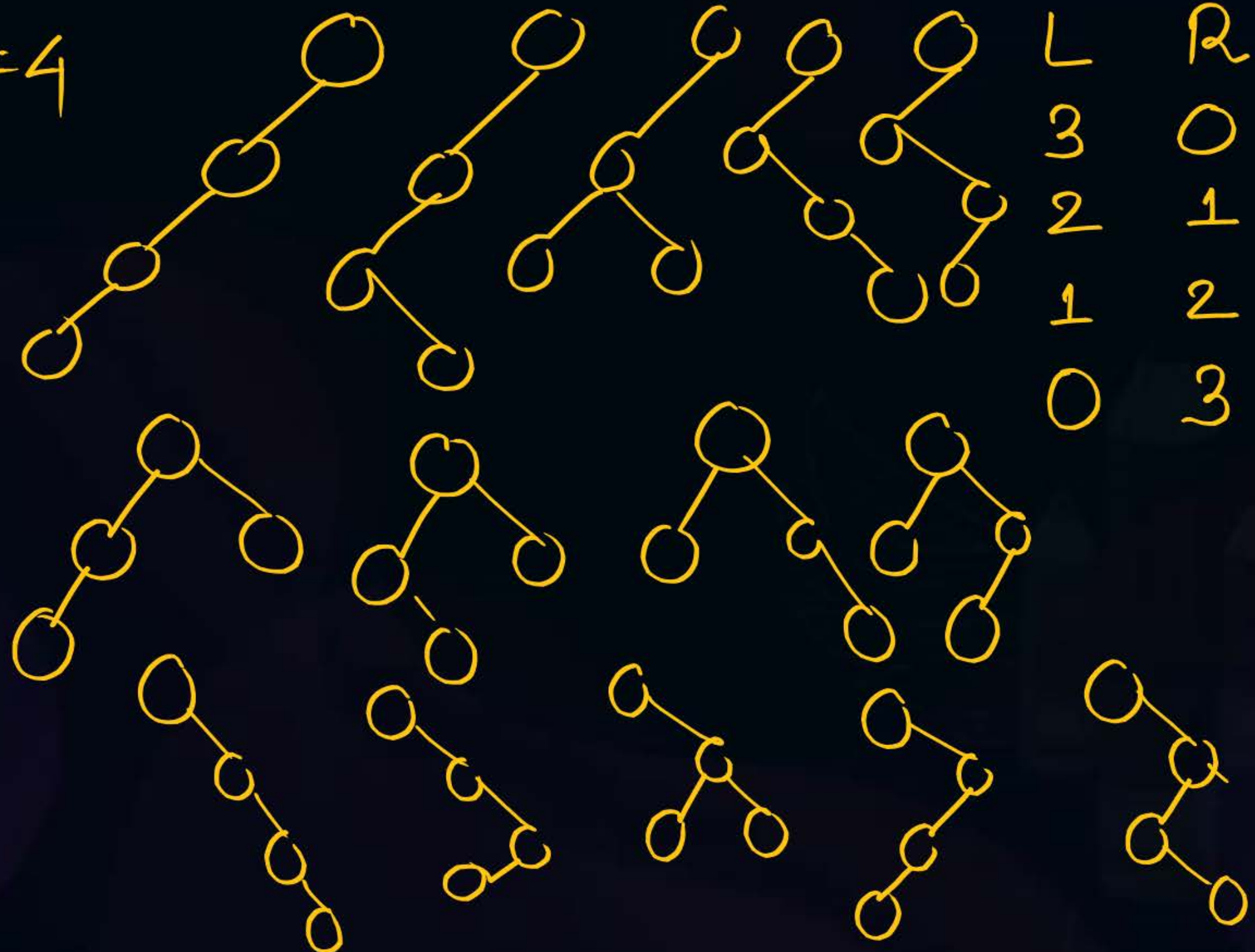
0

2



Topic : Counting of unlabeled Tree

(III) $n=4$



L	R	
3	0	— 5
2	1	— 2
1	2	— 2
0	3	— 5
		14



Topic : Tree



No. of unlabeled tree with n nodes

Catalan No.

$$C_n = \frac{1}{n+1} 2^n C_n$$



Topic : Tree



[B] [C] [D]

$$\underline{L + I = n}$$

#Q Which of the following is FALSE for complete 2-ary tree ?

$$L = I + 1 \Rightarrow \underline{I = L - 1}$$

A If T has I internal nodes then number of nodes is $2I + 1$ ✓ $\leadsto n = I + 1 + I = 2I + 1$

B ✓ If T has L leaves then number of internal nodes is $2L - 1$ False

C ✓ If T has N total number of nodes then number of internal node is $(N + 1) / 2$ $\rightarrow$ false

D ✓ If T has N total number of nodes then number of leaves is $(N - 1) / 2$ False

$$n = 2I + 1 \Rightarrow I = \frac{(n - 1)}{2}$$

$$L + L - 1 = n \quad 2L - 1 = n \quad \leadsto L = \frac{(n + 1)}{2}$$

[MCQ]

[1 Mark]



#Q. Consider a binary tree T in which every node has either zero or two children. Let $n > 0$ be the number of nodes in T. Which ONE of the following is the number of nodes in T that have exactly two children?

Sol. 2 Ig

PYQ. 2025

A $\frac{n-2}{2}$

☒ B $\frac{n-1}{2}$

2 children

C $\frac{n}{2}$

D $\frac{n+1}{2}$

Internal

$I + I + 1 = n$ $\leftarrow$ $L + \underline{I} = n$

$2I = n - 1$

$\underline{I} = \frac{n-1}{2}$

$L = I + 1$



Topic : Question



#Q A 2-3 tree is tree such that

(A) all internal nodes have either 2 or 3 children

(B) all paths from root to the leaves have the same length

if 2-3 tree is having 9 leaf Nodes.

if maximum No. of internal node is $x = 7$

and minimum No. of internal node is $y = 4$

$x + y$ is 11



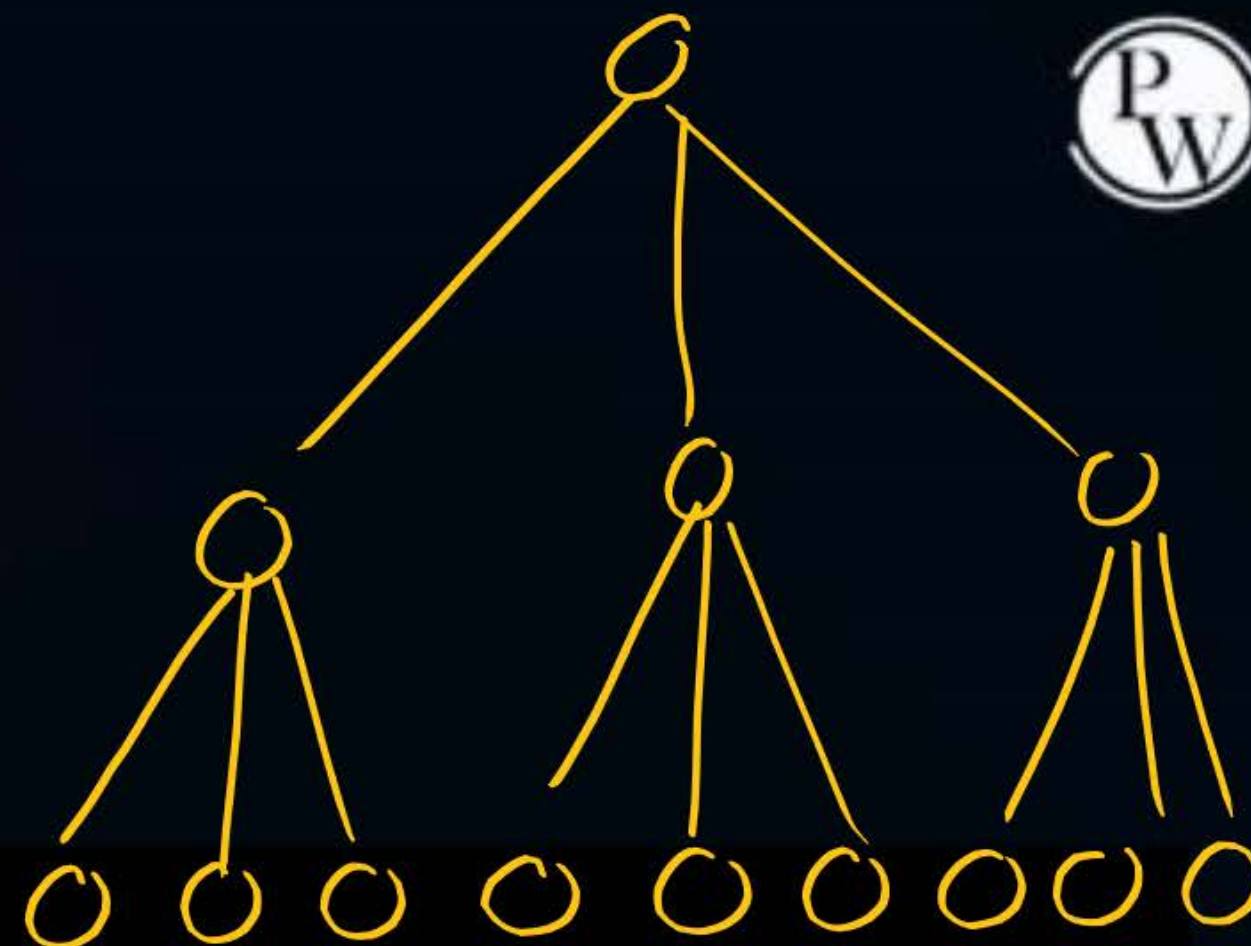
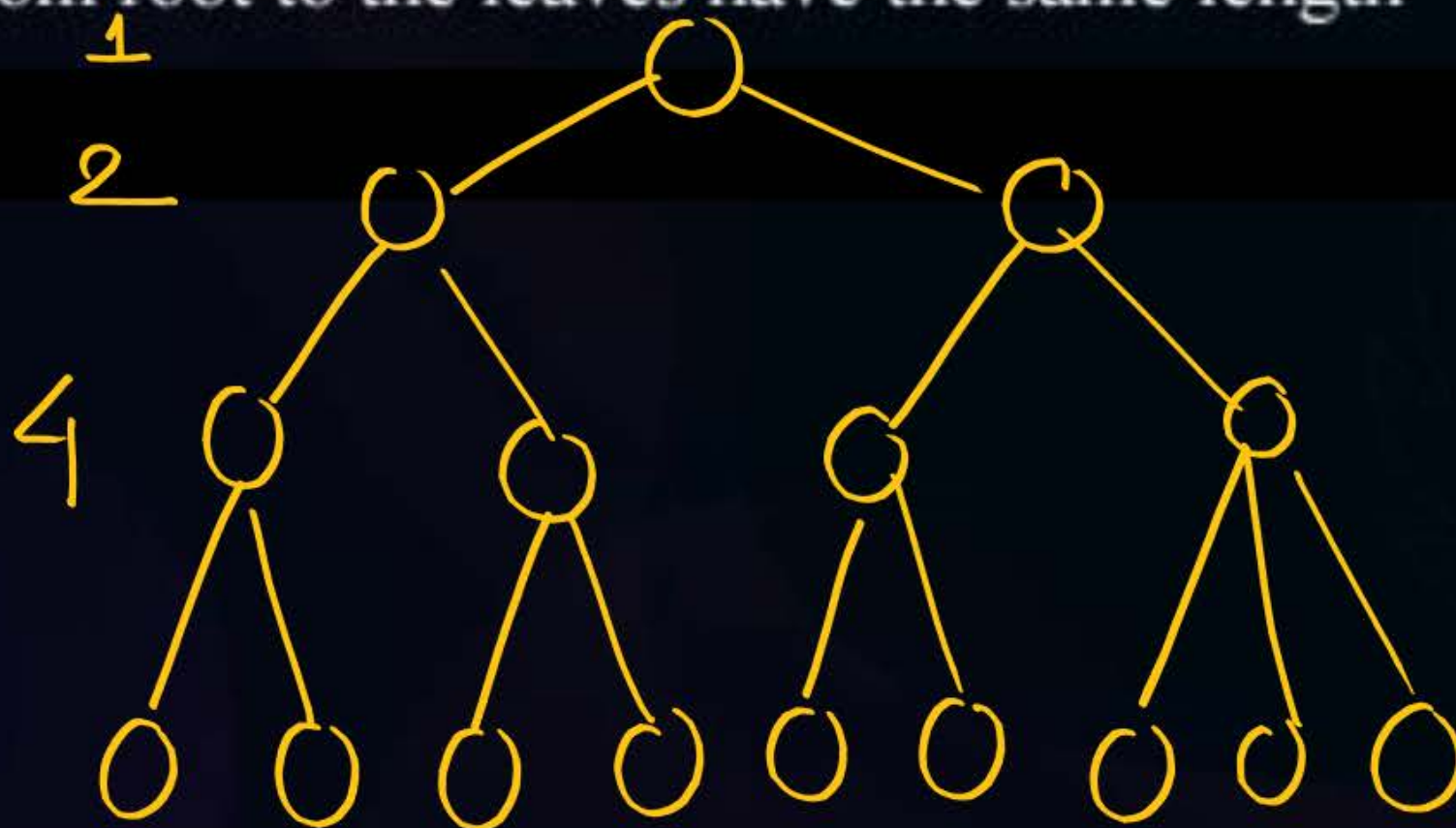
Topic : Question



#Q A 2-3 tree is tree such that

(A) all internal nodes have either 2 or 3 children

(B) all paths from root to the leaves have the same length





Topic : Question



#Q Let T be a binary tree with 127 nodes. The minimum and maximum possible heights of T is x and y then x+y is 6+126 = 132

$$\text{Minimum } \lfloor \log_2 127 \rfloor = 6$$

$$\text{Max } \therefore h = n - 1$$

$$127 - 1 = 126$$



Topic : Question



Consider a binary tree $T1$ that has 200 leaf nodes. The number of nodes in $T1$ that have exactly two children are x and A binary tree $T2$ has 20 leaves, The number of nodes in $T2$ having two children is y then $x+y$ is _____.

$$n_0 = n_2 + 1$$

$$n_2 = n_0 - 1$$

$$x = 200 - 1 = 199$$

$$y = 20 - 1 = 19$$
$$\underline{218}$$



Topic : Question



#Q Let T be a binary tree with 6 leaves and 3 nodes with 1 child then sum of the degrees of all the vertices in T is _____.

degree: Group

$$n_0 = 6$$

$$n_1 = 3$$

$$n_0 = n_2 + 1$$

$$n_2 = 6 - 1 = \underline{5}$$

$$\underline{2 | E |}$$

$$2 \times 13 = \underline{26}$$

total No. of Nodes = 14
Edges = 13



Topic : Question



10 mins - 10es

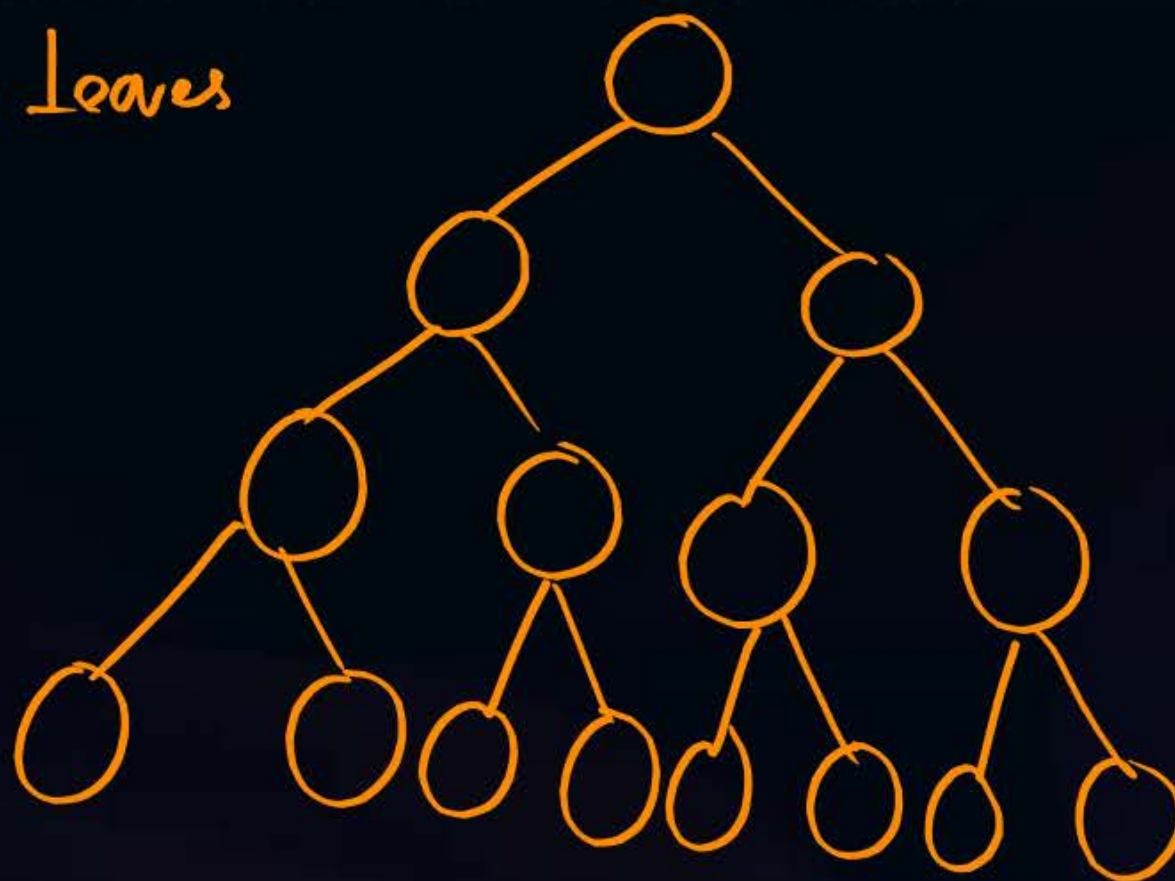
Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.)

Suppose two leaves a and b of T are chosen uniformly and independently at random.

The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) ~~X~~ * 100

No. of ways a, b (two leaves
can be selected)

{ Select a - : 8 ways
Select b - : 8 ways
total = 64 pairs





Topic : Question

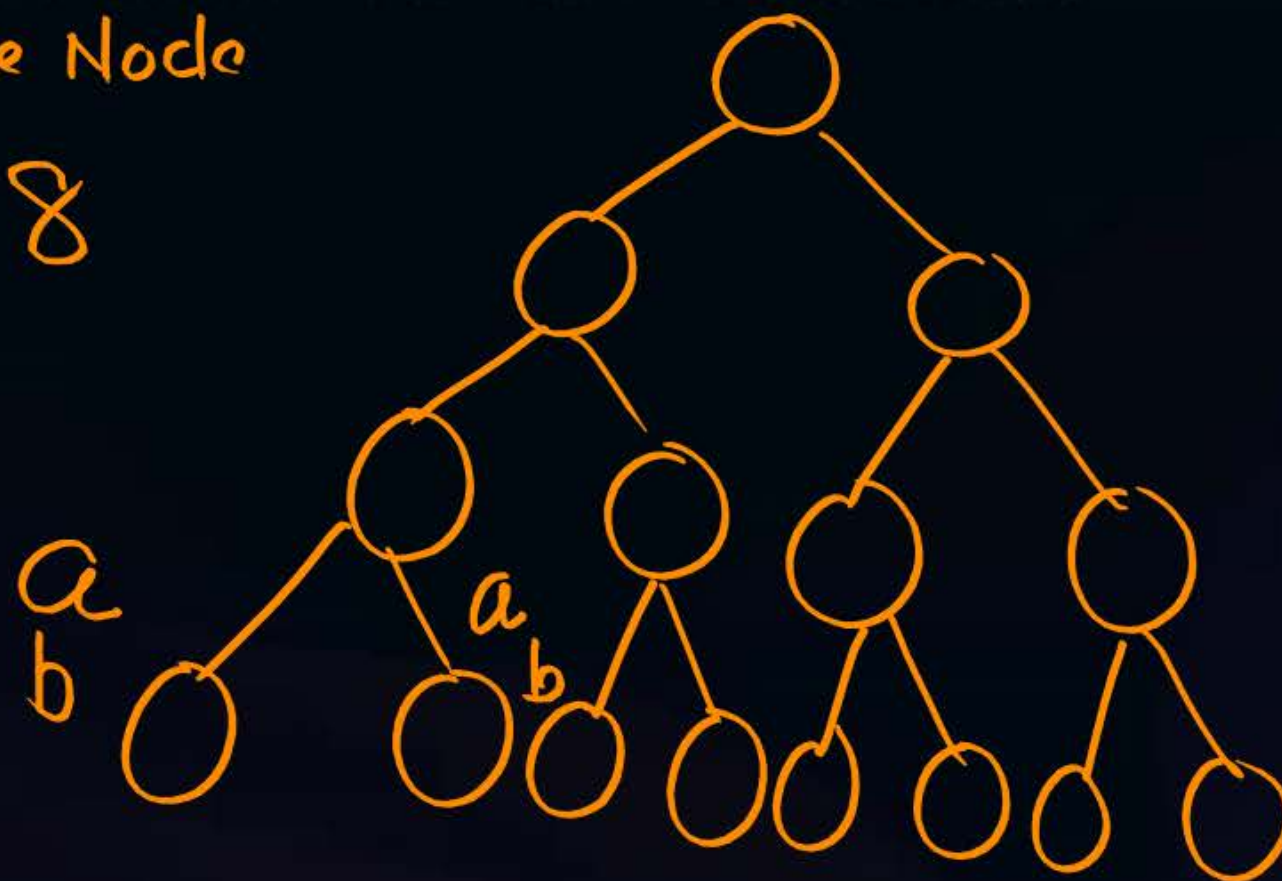


Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.)
Suppose two leaves a and b of T are chosen uniformly and independently at random.
The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) ~~$\frac{100}{8}$~~ $\frac{100}{8}$

I a, b are same Node

0-distance 8

$$\frac{8}{64}$$





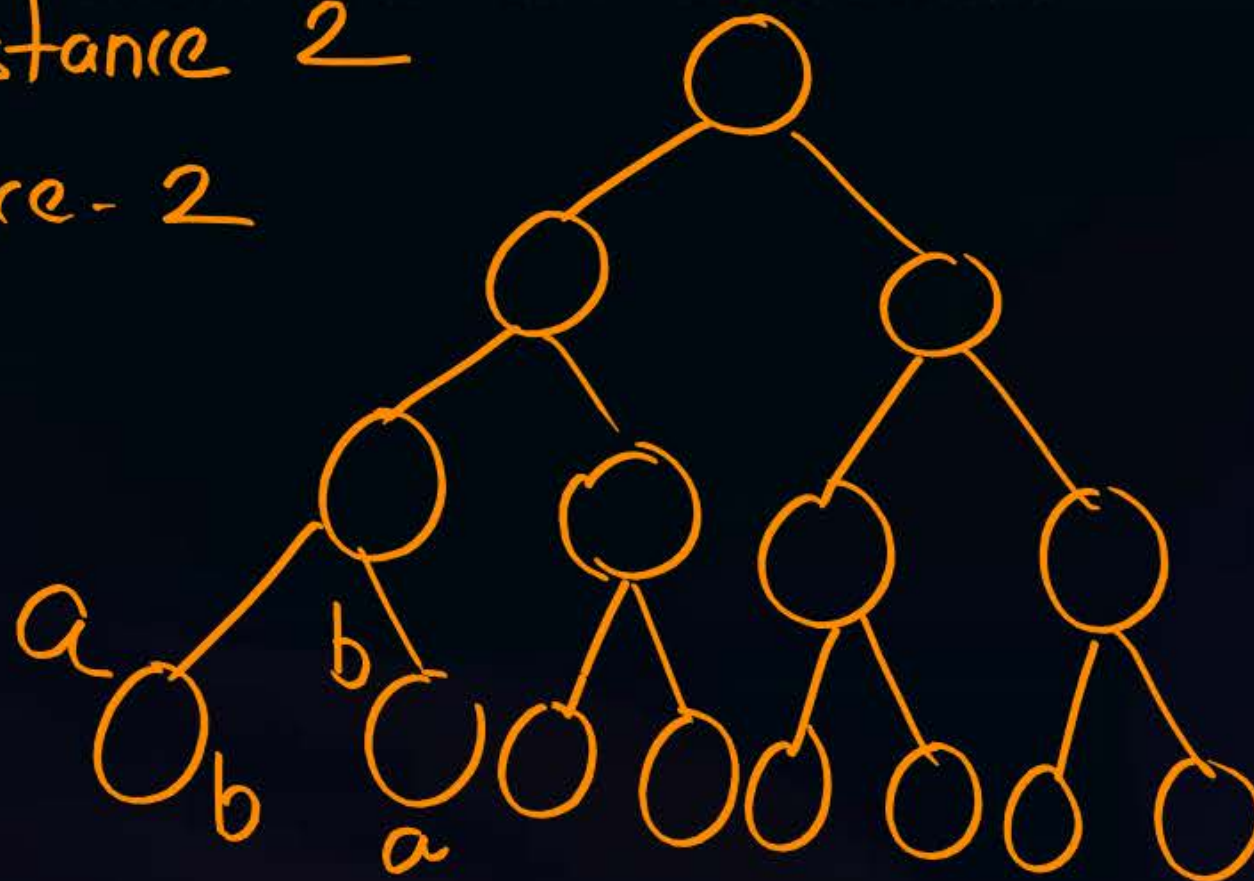
Topic : Question



Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.)
Suppose two leaves a and b of T are chosen uniformly and independently at random.
The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) ~~X~~ * 100

a, b are at distance 2

8 pairs distance - 2





Topic : Question



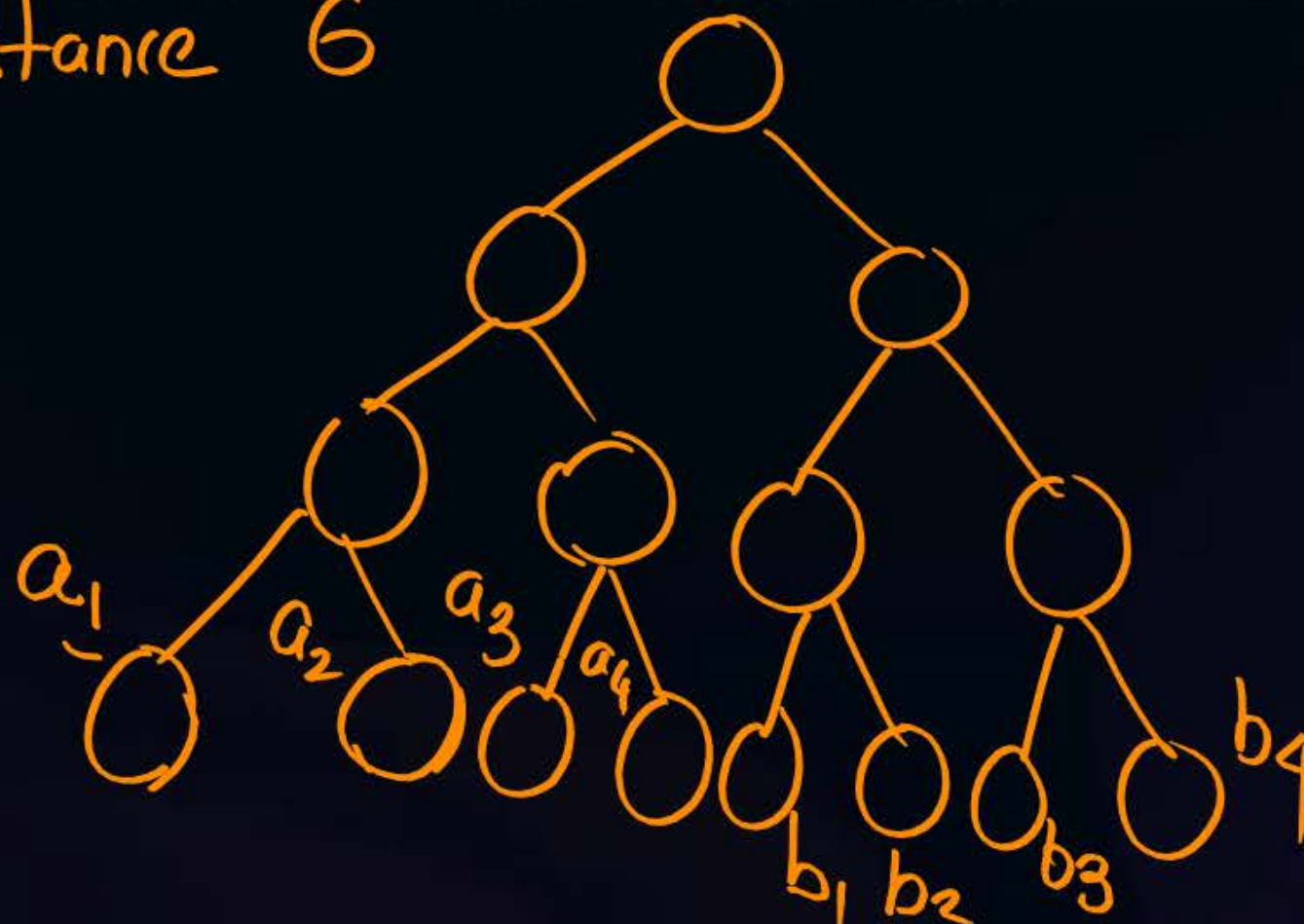
Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.)

Suppose two leaves a and b of T are chosen uniformly and independently at random.

The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) ~~X~~ * 100

a, b are at distance 6

total 32





Topic : Question



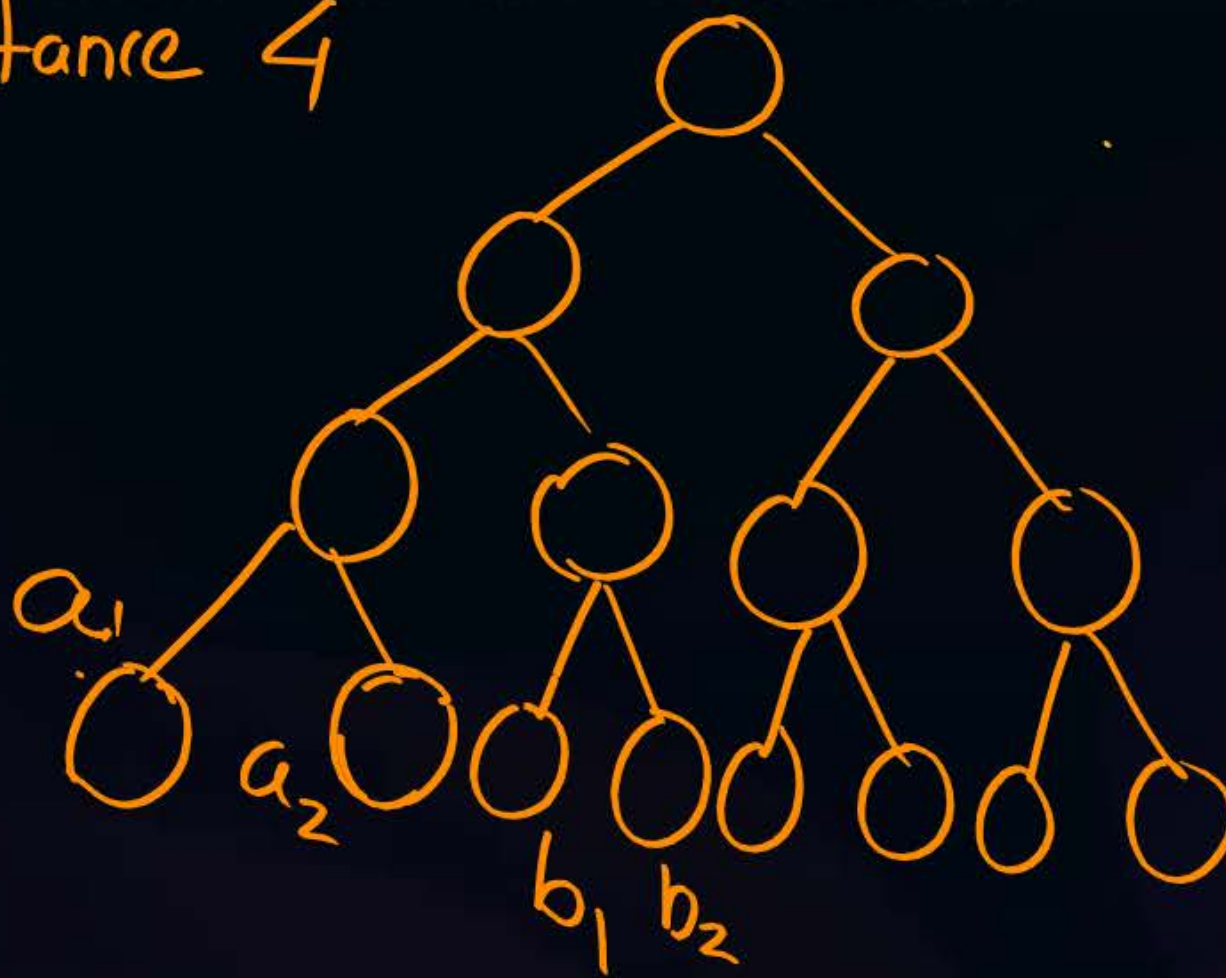
Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.)

Suppose two leaves a and b of T are chosen uniformly and independently at random.

The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) ~~X~~ $\times 100$

a, b are at distance 4

total 16



Expected value $\sum p_i x_i$

$$\frac{8}{64} \times 0 + \frac{8}{64} \times 2 + \frac{16}{64} \times 4 + \frac{32}{64} \times 6$$

$$\left(0 + \frac{2}{8} + 1 + 3 \right)$$

$$0 + 0.25 + 1 + 3 = 4.25 \quad \text{Ans}$$

THANK - YOU